Venkata Lakshmi Parimala Pasupuleti

AI Solutions Engineer · LLM Evaluation & Observability · RAG Pipelines & Agentic Systems · Data Engineering · End-to-End AI Deployment · 9yr Production

+1-703-261-5410 | venkatalakshmiparimalap@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Virginia, USA

PROFESSIONAL SUMMARY

AI solutions engineer with 9 years of production engineering depth and 2 years of hands-on AI system development. I build AI-driven capabilities end-to-end: I designed and built AgentProbe, a production LLM evaluation framework with a 5-metric eval engine, LLM-as-judge scoring, benchmark harness across 1,240 test cases, HITL escalation, and CI-integrated regression gate -- the same evaluation and operationalization rigor this role requires. I built LeaseI, an 8-agent LangGraph AI platform for document intelligence and information extraction from regulatory documents (93.6% precision), demonstrating that AI solutions applied to specialized, large-scale information sources produce reliable, reproducible results when built with the right evaluation discipline. At Entrata, I owned production data pipelines processing millions of compliance-critical records annually and migrated large scale datasets across cloud infrastructure with 100% PHI compliance. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). Portfolio: venkatalakshmiparimala.github.io/Portfolio/about.html

TECHNICAL SKILLS

AI & LLM Systems: LangGraph, Agentic AI, Multi-Agent Orchestration, Large Language Models (LLMs), LLM APIs (Anthropic/Claude, OpenAI, Groq, Ollama), RAG Pipelines, Embeddings, Vector Databases (ChromaDB), Semantic Search, LLM Evaluation, LLM-as-Judge, Hallucination Detection, Benchmark-Driven Evaluation & Experimentation, HITL (Human-in-the-Loop), Agent Observability & Tracing, Prompt Engineering, Output Validation & Guardrails, Document Intelligence, OCR, Information Extraction & Summarization, AI Workflow Automation, AI Chatbots & Conversational Applications, Foundation Model Integration, Model Performance Measurement, Public Trust clearance eligible

Data Engineering & Analytics: Python (9yr production), SQL/PL-SQL, ETL/ELT Pipelines, Data Modeling, PostgreSQL, MySQL, Oracle, Microsoft SQL Server, AWS (S3, EC2, RDS, Glue), Apache Spark (familiarity), Data Governance & Compliance, Data Quality Controls, Large-Scale Dataset Processing, Anomaly Detection (Isolation Forest), Feature Engineering, scikit-learn, Pandas, NumPy; R, Tableau, Power BI

Engineering & Deployment: FastAPI, Flask, REST APIs, React.js, TypeScript, JavaScript, HTML/CSS, Docker, Kubernetes, CI/CD (GitHub Actions), Git, Agile/Scrum; Security & Access Controls, HIPAA-compliant systems, Data Privacy; MLOps practices (model versioning, regression gates, monitoring pipelines)

EXPERIENCE

Magizhs Tech Solution

May 2025 -- Aug 2025

AI/ML Engineer -- Agentic Systems & AI Workflow Automation

- Built an agentic AI mission-planning system orchestrated as a LangGraph stateful graph with conditional routing and guardrails; implemented RAG retrieval over regulatory corpus (FAA Part 107) with cited sources; engineered validation guardrails enforcing compliance constraints before export -- demonstrating the pilot-to-operationalization pattern this role requires.

George Mason University

Aug 2024 -- May 2026

Research Associate & Graduate Teaching Assistant

- Built multi-source prediction models and data pipelines (Python, R) on WHO and UK Road Safety datasets across 190+ countries (R^2=0.977); applied Isolation Forest to automate data integrity checks across 3 research pipelines; designed Tableau and ggplot2 dashboards improving research comprehension 40% for faculty leadership -- demonstrating the scientific data analysis and research pipeline skills federal biomedical work demands.

Entrata -- Multifamily Real Estate SaaS (PropTech)

Sep 2018 -- Aug 2024

Senior Software Engineer -- Data Engineering, Compliance Systems & Cloud Infrastructure · Lehi, UT (Remote)

- **Large-scale data engineering and pipeline ownership** -- sole owner of Entrata's annual HUD income-limit pipeline: federal regulatory data transformed and mapped to ~180,000 domain records powering eligibility and rent calculations for 100K+ housing units. Defined data transformation rules, validation logic, and quality controls; conducted pre-release data audits; managed reconciliation and rollback. Zero data integrity violations across 7 years. This is the production data engineering discipline that operationalizing AI on large, complex information sources requires.
- **Scalable, compliant cloud data infrastructure** -- led AWS S3 migration with Libsodium-encrypted ETL pipelines (100% PHI compliance, zero security incidents); RabbitMQ-based async processing with idempotency guarantees; optimized 200+ SQL queries (30% performance gain); Docker microservices redesign (50% scalability improvement). Demonstrated cloud-based and containerized environments for scalable data workloads.
- **Cross-functional collaboration and requirements translation** -- partnered with operations, compliance, and business stakeholders to translate requirements into technical solutions across 120+ Agile sprints; 80% manual workload eliminated through automation pipelines; mentored 5+ engineers; 300+ peer code reviews; 99.5% uptime.

Entrata India -- Pune

Jan 2017 -- Sep 2018

Associate Software Engineer -- Data Pipelines & Compliance Workflows

- Built compliance data pipelines (income-limit import, 1.6M HUD/Tax Credit records); AWS S3 migration; database schema redesign with post-deployment stabilization.

PROJECTS

AgentProbe: Production LLM Evaluation Framework -- Benchmark-Driven & HITL

LLM Evaluation · RAG · Embeddings · ChromaDB · LLM-as-Judge · HITL · CI/CD

Python · LangGraph · FastAPI (15-endpoint REST API) · ChromaDB · React.js · TypeScript · GitHub Actions | github.com/venkatalakshmiparimala/AgentProbe

Production LLM evaluation and observability framework built for operationalizing AI solutions: 5-metric eval engine (correctness, groundedness, hallucination rate, latency, token efficiency), LLM-as-judge scoring with multi-trial pass@k harness, ChromaDB vector evaluation, CI-integrated regression gate blocking quality degradation, structured tracing for failure analysis, multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama). Used to evaluate LeaseI across 1,240+ test cases. Directly demonstrates the "evaluate, iterate, and operationalize AI solutions" capability this role requires.

LeaseI: AI Platform for Document Intelligence & Information Extraction

LangGraph · RAG · Information Extraction · Agentic AI · HITL · Foundation Models

Python · LangGraph (8-agent) · FastAPI · TF-IDF RAG · OCR · SQLite · GitHub Actions | github.com/venkatalakshmiparimala/Lease_complaince_rag

End-to-end AI solutions engineering: designed, built, and evaluated an agentic AI platform for document intelligence and information extraction from large-scale specialized documents (1,050 synthetic tenants, 5 US states + federal regulatory corpus). OCR-fallback ingestion, section-aware chunking, TF-IDF RAG, 8-agent LangGraph pipeline with conditional routing and HITL escalation, AI chatbot for self-service queries with safety boundaries. 93.6% precision, 87.9% F1. The same end-to-end methodology applies directly to biomedical literature, grants, or policy documents.

AI on FHIR: Secure Health Data Platform -- HIPAA & Federal Compliance

Flask · REST API · HIPAA · RBAC · NLP · Federal Data Standards

Python · Flask · React · Docker · FHIR API · spaCy · BERT | github.com/venkatalakshmiparimala/AI_on_FHIR

HIPAA-compliant AI platform integrating NLP (BERT, spaCy) with FHIR-standard federal health data systems; RBAC access controls; secure data retrieval across structured clinical records. Demonstrates AI applied to federal health data standards and the data governance, security, and compliance considerations this federal contractor role requires.

EDUCATION

M.S. Data Analytics Engineering -- George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering -- JNTU Anantapur — 2013--2017